Continuous functions on the Cauchy reals agree if they agree on the rationals

The Univalent Foundations Program

hott-book:reals.tex:local-snapshot-20261011; CC BY-SA 3.0 Unported

Editorial note. The counted statement is the author's 'RC-continuous-eq' (source lines 1372-1378, Lemma 11.3.39); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. The retained counter values (chapter 11, section 3, theorem offset 38) make the counted lemma print as Lemma 11.3.39, matching the original book numbering taken from the label-number dump of the pinned snapshot.

Editorial note. The single external reference is the cited closeness statement invoked at source line 1390; it is a dump-confirmed navigation pointer and does not replace any part of the counted proof, which is retained in full.

Editorial note. The retained preamble is the chapter's own retained starter preamble for this pinned source.

Editorial scope. One original result of the HoTT Book, counted once: continuous functions agreeing on rationals are equal (Lemma 11.3.39).

Allowed standard prerequisites (editorial).

- Continuous function on the Cauchy reals. f is continuous when for every u and epsilon there is delta with v close-delta u implying f u close-epsilon f v.. It supplies the two moduli theta and eta at source lines 1385-1388.
- Induction on the Cauchy reals. RC is generated by rational points and limits of Cauchy approximations.. It is the induction performed at source line 1380.
- Closeness of limits to the approximating sequence. lim(x) is close to x_delta for small delta.. It is the retained cited result used at source line 1390 and it is what supplies v.

Source reference. Original lemma 11.3.37: sources/hott-reals/book-labelnumbers-first-edition.tex, source lines 1871–1871. Original result cited by the retained text but located elsewhere in the chapter. It is a navigation pointer only: the counted statement's own author proof is retained line for line and nothing is replaced by it. The number is taken from the label-number dump of the pinned book snapshot. It is kept as a pointer rather than as a context span because retaining it as context would either print it with the wrong section number or consume a number in the retained sequence.

Lemma 11.3.39. *Let $f, g : \mathbb{R}_{\mathsf{c}} \to \mathbb{R}_{\mathsf{c}}$ be continuous, in the sense that*

$$\forall(u : \mathbb{R}_{\mathsf{c}}).\, \forall(\epsilon : \mathbb{Q}_+).\, \exists(\delta : \mathbb{Q}_+).\, \forall(v : \mathbb{R}_{\mathsf{c}}).\, (u \sim_\delta v) \to (f(u) \sim_\epsilon f(v))$$

and analogously for g. If $f(\mathsf{rat}(q)) = g(\mathsf{rat}(q))$ for all $q : \mathbb{Q}$, then $f = g$.

Proof. We prove $f(u) = g(u)$ for all u by $\mathbb{R}_{\mathsf{c}}$-induction. The rational case is just the hypothesis. Thus, suppose $f(x_\delta) = g(x_\delta)$ for all δ. We will show that $f(\mathsf{lim}(x)) \sim_\epsilon g(\mathsf{lim}(x))$ for all ϵ, so that the path constructor of $\mathbb{R}_{\mathsf{c}}$ applies.

Since f and g are continuous, there exist θ, η such that for all v, we have

$$(\mathsf{lim}(x) \sim_\theta v) \to (f(\mathsf{lim}(x)) \sim_{\epsilon/2} f(v))$$
$$(\mathsf{lim}(x) \sim_\eta v) \to (g(\mathsf{lim}(x)) \sim_{\epsilon/2} g(v)).$$

Choosing $\delta < \min(\theta, \eta)$, by original lemma 11.3.37 we have both $\mathsf{lim}(x) \sim_\theta y_\delta$ and $\mathsf{lim}(x) \sim_\eta y_\delta$. Hence

$$f(\mathsf{lim}(x)) \sim_{\epsilon/2} f(y_\delta) = g(y_\delta) \sim_{\epsilon/2} g(\mathsf{lim}(x))$$

and thus $f(\mathsf{lim}(x)) \sim_\epsilon g(\mathsf{lim}(x))$ by the triangle inequality. □